

MEMORANDUM FOR: DFAS Cleveland Personnel

SUBJECT: DFAS Cleveland Safety Office Industrial Hygiene
Report: March 23, 2018 – Floors [REDACTED]
[REDACTED]

Attached please find the latest environmental tests conducted for DFAS here at the [REDACTED]. For the CONCLUSIONS and RECOMMENDATIONS, please see pages 9 through 12 of this report.

Please direct any questions to the Safety Office @
[REDACTED]

DFAS Safety Office

[REDACTED]
[REDACTED]
[REDACTED]
March 23, 2018

[REDACTED]
[REDACTED]
[REDACTED]
Re: Report regarding general indoor air quality (IAQ) parameter testing, fungal air sampling, airborne fiber sampling, ambient particulate testing and visual evaluation performed for the DFAS occupied areas within [REDACTED] in the [REDACTED] located at [REDACTED] Cleveland, Ohio [REDACTED]

Dear [REDACTED]

INTRODUCTION

On February 27 and March 1, 2018, [REDACTED] an industrial hygienist representing [REDACTED] was on-site performing a general IAQ screening which included measuring the temperature, relative humidity, carbon dioxide, and carbon monoxide levels, microbial air sampling, airborne fiber sampling, ambient particulate testing and cursory visual inspections of the [REDACTED] floors and the sub basement training room. [REDACTED] scope of work included performing the Phase 1 IAQ screening within the DFAS occupied areas located on the [REDACTED] selected floors and areas in the [REDACTED] located at [REDACTED] in Cleveland, Ohio. The Phase 1 IAQ screening was being performed to identify possible sources/conditions, which may be deemed an IAQ concern. No other environmental issues, besides those previously identified, were addressed during the evaluation such as lead, chemical exposure, etc.

DFAS was located on [REDACTED] within the [REDACTED] however, only the [REDACTED] floors and the sub basement training room were in [REDACTED] scope of work. The interior spaces on the floors [REDACTED] At the time of the site visit the structure was operational and occupied. The site visit and IAQ evaluations were part of annual periodic monitoring.

The interior building materials consisted of carpet or floor tile on a concrete floor slab, plaster or drywall walls with associated joint compound and either metal or lay-in ceiling tiles. The duct, pipe and electrical systems were located above the ceiling tiles in the ceiling plenum space. All floors contain sprayed-on fireproofing in the ceiling space however, this material was not observed or evaluated. The individual areas had typical furnishings associated with each service that was provided.

[REDACTED]
[REDACTED]

METHODOLOGIES

Indoor Air Quality Screening

Components of the screening of the indoor environment included measurements of temperature, relative humidity (RH), carbon dioxide (CO₂), carbon monoxide (CO) concentrations and airborne ambient particulate testing. The technical approach used for the IAQ evaluation followed standard IAQ evaluation protocols.

Temperature and Relative Humidity

CTG conducted general IAQ parameter testing throughout the accessible office spaces and outside the building. Temperature and relative humidity measurements were collected at each test location. Measurements were collected with a direct reading hygrometer Q-Trak 7565.

The American Society of Heating, Refrigeration and Air Conditioning (ASHRAE) recommends that indoor temperatures be maintained between 68°F and 76°F during the winter/transitional (spring) season and between 74°F and 78°F during the summer/transitional (fall) season with relative humidity between 30% and 60%. These values are considered acceptable ranges of temperature and humidity for sedentary persons wearing typical summer clothing and typical winter clothing. These "comfort zones" assume that occupants are engaged in only light activity, such as what would be found in a typical office setting.

The ideal comfortable relative humidity range has been reported as 40% to 60%, as long as building materials or contents are not adversely affected. Low relative humidity can result in eye irritation and complaints of nose and throat discomfort. High humidity levels (over 70%) can promote the growth of microorganisms on building surfaces and furnishings, and cause or contribute to microbial IAQ problems.

Carbon Dioxide (CO₂)

Carbon dioxide measurements were collected with a Q-Trak 7565 direct reading CO₂ monitor. General industry standards indicate that as CO₂ levels exceed 1,000 ppm, some occupants may complain of stagnant air, stuffy conditions and poor ventilation.

Carbon dioxide, a product of combustion and human respiration, is a commonly used indicator of overall air quality and ventilation rates within an occupied building. The levels found in buildings are primarily a function of the rate and amount of outside air delivery to the occupied space, the effectiveness of air distribution within the space, and the occupancy of the space.

It is generally accepted that an inadequate fresh air supply may cause problems such as irritation of the eyes, nose, and throat; headaches; and respiratory difficulties due to a general build up of non-specific contaminants and odors. These symptoms are commonly categorized as Sick Building Syndrome (SBS).

Carbon Monoxide (CO)

Carbon monoxide, typically a product of incomplete combustion, is a chemical asphyxiate that can cause headaches in low concentrations. The EPA National Ambient Air Quality Standard for CO of 9 ppm, as an eight-hour average, and 35 ppm as a one-hour average, has been cited by ASHRAE as guidelines for IAQ. Carbon monoxide measurements were collected with a Q-Trak 7565 direct reading CO meter.

Airborne Ambient Particulate Sampling (dust)

The technical approach used for the particulate sampling followed standard industrial hygiene protocols for the parameters included in the scope of work. The particulate testing was performed with a P-Trak Ultrafine Particulate Counter 8525 (PTrak). The PTrak can measure very small particulate. The 50% cut-off for small particles is 0.02 micrometers (μm), which means that half of the 0.02 μm particles are counted and half are not. As soon as the particle reaches 0.04 μm , almost 100% are counted. The maximum particle size is 1.0 μm . Few particles larger than 1.0 μm are counted by the instrument because they impact and stick inside the switching valve of the instrument. Statistically speaking, there are few particles above 0.1 μm in normal ambient air. The percent of particles is called the "counting efficiency". The PTrak is based on a miniature, continuous flow Condensation Nucleus Counter (CNC), also known as a Condensation Particulate Counter (CPC). A CNC takes particles that are too small to be easily detected, grows them to a larger, easily detectable size and then counts them. The PTrak grows submicron particles to submicron alcohol droplets and then measures the concentration of the alcohol droplets. This makes the PTrak sensitive to particles having diameters as small as 0.02 microns, but sensitive to variations in particle size, shape, composition and refractive index. The PTrak is a hand held portable version of the PortaCount.

Microbial Air Sampling Technical Approach

The technical approach used for the microbial sampling followed standard industrial hygiene protocols for the parameters included in the scope of work. The IAQ protocols and assessment criteria were based on those established by the NIOSH; the Environmental Protection Agency (EPA); the American Industrial Hygiene Association (AIHA); the American Conference of Governmental Industrial Hygienists (ACGIH); and information in the scientific literature.

Microbial air sampling was performed inside within DFAS occupied areas located on the [REDACTED] floors, the sub basement training room and outside. Countable spore samples were collected on Allergenco-D air-sampling cassettes in line with a high volume air sampling pump. The Allergenco-D cassettes are manufactured by Environmental Monitoring Systems. The high volume air pump was calibrated in accordance with standard industrial hygiene practices at 15.0 liters per minute (L/min). Sampling time for the spore trap was six (6) minutes for a possible detection limit of 11 spores per cubic meter (spores/m³) of air. The Allergenco-D cassette collects the airborne aerosols on a glass slide contained in the cassette housing. The glass slides are analyzed by light microscopy in accordance with the manufacturer's directions. Presumptive species and particle identification is based on morphology and is performed in accordance with information provided in standard reference manuals. Positive identification of spore types is not always possible for some species. The air samples were analyzed to measure the total concentration of countable fungal spores in spores/m³ and to identify each of the spore types and particles and their concentrations in spores/m³.

Direct Microscopic Examination Samples

The samples were collected using the previously described techniques. [REDACTED] analyzed the samples. The samples were analyzed to determine the presence of fungal spores and particle characterization by direct microscope examination under a light microscope. Particle identification of the fungal spores and structures was then performed in accordance with information provided in standard reference manuals. Positive identification of spore types is not always possible e.g., under the microscope *Penicillium* and *Aspergillus* spores cannot be differentiated; therefore, the laboratory reports the presence of such spores as the *Aspergillus/Penicillium* group.

Airborne Fibers

The air sampling for fibers was performed to determine fiber concentrations of the DFAS occupied areas/rooms on the 9 floors. [REDACTED] industrial hygienist collected ambient air samples and field blanks that were analyzed by Phase Contrast Microscopy (PCM). The PCM air samples were collected by standard industrial hygiene practices. Air was drawn through 25-millimeter (mm) diameter, 3 piece cassettes with 50 mm extension cowls made from non-conductive, carbon-filled polypropylene. The filter was made from 0.8 micrometer porosity, mixed cellulose ester (MCE), supported by a cellulose pad fitted into the base section of the cassette. High volume air sampling pumps with variable flow adjusters were utilized for sample collection. The air sampling pumps were calibrated prior to and following the air sampling with a rotometer.

The PCM samples were submitted to [REDACTED] for fibers per field analysis by PCM NIOSH 7400. [REDACTED] is accredited by the National Institute of Standards of Technology (NIST) through the National Voluntary Accreditation Program (NVLAP), license number [REDACTED] for asbestos analysis.

RESULTS

Temperature/Relative Humidity/Carbon Dioxide (CO₂)/Carbon Monoxide (CO)/Particulate

████████ collected data regarding temperature, relative humidity, carbon dioxide and carbon monoxide throughout DFAS occupied rooms and areas located on the ██████████ floors, the sub basement training room within the ██████████ and outside. The results and locations tested are listed in the following chart:

Station ID	Location	Time	Temp (F)	RH (%)	CO ₂ (ppm)	CO (ppm)	Particulate (p/cm ³)
1	████████ Floor	10:26 A	72.4	26.1	775	0.0	393
		1:08 P/4:53 P	75.6	24.7	726	2.1	563
2	████████ Floor	10:32 A	74.3	26.0	950	0.2	633
		1:12 P/4:56 P	75.8	24.8	851	2.1	766
3	████████ Floor	10:36A	73.2	25.7	792	0.0	1080
		1:14 P	75.6	24.7	X	X	1020
4	████████ Floor	10:44 A	75.5	25.5	791	0.1	932
		1:16 P	76.3	24.5	X	X	2400
5	████████ Floor	10:48 A	76.0	25.2	822	0.1	380
		1:18 P	76.5	24.6	X	X	515
6	████████ Floor	10:53 A	75.4	25.1	730	0.2	679
		1:22 P	76.9	24.5	789	1.9	782
7	████████ Floor	11:05 A/10:08 A	76.2	24.9	893	2.1	796
		1:25 P/5:00 P	77.1	24.9	695	1.8	2920
8	████████	11:09 A/10:10 A	76.4	24.8	842	1.9	486
		1:27 P	77.4	24.2	X	X	933
9	████████ floor	11:12 A/10:13 A	75.8	24.8	837	1.8	783
		1:28 P	77.6	24.2	X	X	981
10	████████ Floor	11:16 A/10:14 A	75.4	25.1	802	1.7	849
		1:30 P	76.9	24.3	X	X	1600
11	████████	11:30 A/10:16 A	72.1	26.0	794	1.6	700
		1:35 P/5:03 P	76.2	24.6	655	1.6	953
12	████████ Floor	11:32 A/10:17 A	73.9	25.7	863	1.7	688
		1:37 P	76.8	24.5	X	X	1080
13	████████ Floor	11:34 A/10:18 A	75.1	25.0	780	1.7	957
		1:39 P	76.5	24.3	X	X	1520

██████████, Cleveland, Ohio
Indoor Air Quality Evaluation

Station ID	Location	Time	Temp (F)	RH (%)	CO2 (ppm)	CO (ppm)	Particulate (p/cm ³)
14	██████████ Floor ██████████	11:37 A/10:19 A	75.8	25.0	849	1.6	729
		1:41 P	76.4	24.5	X	X	1040
15	██████████ Floor ██████████ doorway/hall	11:46 A/10:24 A	76.1	25.1	798	1.6	330
		1:46 P/5:07 P	75.8	24.8	690	1.6	399
16	██████████ Floor ██████████ /40	11:49 A/10:24 A	76.4	24.6	851	0.8	4400
		1:48 P	76.2	24.7	X	X	2390
17	██████████ Floor ██████████ /194-47/48	12:03 P/10:26 A	76.6	24.6	800	1.5	755
		1:50 P	76.2	24.5	X	X	912
18	██████████ Floor ██████████ /hallway	12:13 P/10:30 A	76.4	24.5	695	1.5	671
		1:54 P/5:10 P	76.6	24.4	620	1.5	942
19	██████████ Floor ██████████	12:18 p/10:33 A	75.8	24.7	709	1.5	1400
		1:58 P/5:12 P	77.1	24.2	610	1.5	6630
20	Outside – west entrance	10:10 A/10:02 A	55.4	28.9	375	0.5	7320
		2:16 P/4:48 P	69.5	25.5	407/381	1.7/2.3	11,700
21	Sub basement training room	12:54 P	73.9	25.6	Locked	Locked	1790
		2:11 P	74.0	25.1	Locked	Locked	6000

ASHARA Ranges

Temperature – 68°F and 76°F during the winter/transitional (spring) season

RH – 40-60%

CO₂ – ≤1,000 ppm

CO – 9 (8 hour period of time), 35 (1 hour period of time)

Sample #	Sample Location	Sample Type	Result (f/cc*)
044 22718	22	Ambient	<0.005
044 22718	23	Ambient	<0.005
044 22718	24	Ambient	<0.005
044 22718	25	Ambient	<0.005
044 3118	26	Ambient	<0.004
044 3118	27	Ambient	<0.004
044 3118	28	Ambient	<0.004
044 3118	29	Ambient	<0.003
044 3118	30	Ambient	<0.005
044 3118	31	Ambient	<0.005
044 3118	32	Ambient	<0.005
044 3118	33	Ambient	Pump was turned off
044 3118	34	Ambient	<0.005
044 3118	35	Blank	1 Fiber/100 Fields

Microbial Air Sampling

Sample #		Location	Spores Identified	Results (spores/m ³)
044 3118	36	██████████ – ██████████	Ascospores	11
			Aspergillus/Penicillium	22
			Basidiospores	11
			Cladosporium species	11
			Total	56
044 3118	37	████ Floor – column between ████████████████████	Ascospores	11
			Basidiospores	11
			Total	22

██████████, Cleveland, Ohio
Indoor Air Quality Evaluation

Sample #		Location	Spores Identified	Results (spores/m ³)
044 3118	38	████ Floor – █████	Aspergillus/Penicillium	11
			Total	11
044 3118	39	████ Floor – █████ doorway	Aspergillus/Penicillium	11
			Total	11
044 3118	40	████ Floor – █████	Aspergillus/Penicillium	11
			Basidiospores	11
			Total	22
044 3118	41	████ Floor – █████ hallway corner	Ascospores	11
			Aspergillus/Penicillium	22
			Basidiospores	11
			Total	44
044 3118	42	████ Floor – █████	Ascospores	11
			Basidiospores	11
			Total	22
044 3118	43	████ Floor – column adjacent west █████	Ascospores	11
			Aspergillus/Penicillium	11
			Basidiospores	22
			Epicoccum species	11
			Total	56
044 3118	44	████ Floor – column by █████	Basidiospores	11
			Cladosporium species	22
			Total	33
044 3118	45	████ Floor – █████ cubicle	Basidiospores	22
			Total	22
044 3118	46	████ Floor – █████	Basidiospores	11
			Total	11
044 3118	47	████ Floor – column adjacent east of █████	Ascospores	11
			Basidiospores	22
			Total	33

Sample #		Location	Spores Identified	Results (spores/m ³)
044 3118	48	Sub basement – training room's NW corner	Ascospores	56
			Aspergillus/Penicillium	11
			Basidiospores	22
			Chaetomium species	11
			Cladosporium species	11
			Smuts/Myxomycetes	11
			Total	122
044 3118	49	Outside – west entrance	Ascospores	111
			Basidiospores	33
			Total	144
044 3118	50	Lab blank	None detected	

Visual Observations

At the time of the site visits on February 27 and March 1, 2018, it was approximately 57°F and partly sunny on February 27th during the Temp/RH/CO/CO₂ sampling and 30°F and rainy/snowy on March 31st during the microbial sampling. ████████ did not detect any musty smells or unusual odors throughout the DFAS occupied rooms and areas located on the ████████ floors, the sub basement training room within the ████████. No areas of moisture, staining or suspect mold growth were observed in any of the rooms and areas of the DFAS occupied evaluated floors. During the visual evaluation, the structure was occupied and operational. The HVAC system was operating in the heating mode.

CONCLUSIONS

Temperature

The results of the temperature readings in the DFAS occupied areas of the ████████, ████████ floors, and the sub basement training room within the ████████ ranged from 72.1 to 77.6 degrees F, which is slightly above the generally accepted ASHRAE guidelines of 68°F to 76°F during the winter to spring transition season in some areas. The slightly elevated temperatures were measured during some of the morning readings on some floors and areas and began to increase slightly in the afternoon within the majority of the floors/areas.

Relative Humidity

The results of the relative humidity testing in the DFAS occupied areas of the [REDACTED]¹, [REDACTED] floors, and the sub basement training room within the [REDACTED] were 24.2% to 26.1%. The generally accepted ASHARA relative humidity range is 40-60%, which is considered a proper comfort zone. All of the areas had readings that were lower than the generally acceptable "comfort zone" range. Low relative humidity can result in eye irritation and complaints of nose and throat discomfort.

Carbon Dioxide (CO₂)

The results of the carbon dioxide ranged from 610 ppm to 950 ppm in the occupied areas of the [REDACTED], [REDACTED] floors, and the sub basement training room within the [REDACTED] at the time of the testing. The results were less than the general industry standard concentration of 1,000 ppm. The general industry standard indicates that readings at or below 1,000 ppm level indicates satisfactory air exchanges and input of outside air (in a fully occupied building).

Carbon Monoxide (CO)

The results of the CO testing was between 0.0 ppm and 2.1 ppm at all the testing locations throughout the occupied areas of the [REDACTED] floors, and the sub basement training room within the [REDACTED] at the time of testing. The EPA National Ambient Air Quality Standard for CO of 9 ppm, as an eight-hour average, and 35 ppm as a one-hour average, has been cited by ASHRAE as guidelines for IAQ.

Airborne Ambient Particulate (dust)

The particulate testing was performed in the locations that the other general IAQ parameters were measured. Each reading tested the air for 10 seconds at each location. The testing was performed over the course of approximately 3.5 hours on February 27th. The results listed in the chart are presented in particles per centimeter cubed (p/cm³) of air. The lowest reading was 393 p/cm³ and the highest reading was 6,000 p/cm³. The lowest outdoor ground level reading was 6,630 p/cm³ and the highest reading was 11,000 p/cm³. The results indoors were significantly lower than outside.

Airborne Fibers

Based upon [REDACTED] airborne fiber sampling conducted on February 27th and March 1st, 2018, the levels of fibers in air were below the asbestos standard of less than or equal to 0.01f/cc when using the "final clearance" standard as a comparative.

Microbial Air Sampling

The results of the microbial air samples collected on March 1, 2018, indicated that the total numbers of microbial spores were lower inside the DFAS occupied areas on the 3 ██████████ floors, and the sub basement training room within the ██████████ when compared with the outside environment, indicating that there was not a source of microbial growth impacting the ambient conditions associated with the indoor environment at the time of the sampling. The individual species inside, that were also identified outside, were at lower levels inside. The individual species of microbial spores: Aspergillus/Penicillium, Chaetomium species, Cladosporium species, Epicoccum species and Smuts/Myxomycetes were identified at low levels inside and not outside. Four of these species are from plants and their spore counts were very low. The seven areas where Aspergillus/Penicillium was identified contained only 1 or 2 spores in areas: 1 spore in 285-47 (28th), 2669D (26th), 252-31 (25th), 242-38 (24th) and SB29 training room and 2 spores in 30A-17/18 (30th) and 25A-7/8 (25th). These are also very low spore counts.

As a general rule, the normal closed-mode indoor total fungal spore counts across the central portion of the U.S. (bounded by a latitude of, say, 35 degrees north to 45 degrees north), for healthy buildings (buildings not experiencing fungal problems) is usually less than 500 counts per cubic meter of air (counts/m³).

RECOMMENDATIONS

The conditions observed on site indicate that the IAQ inside the DFAS occupied areas on the ██████████ floors, and the sub basement training room within the ██████████ at the time of the sampling and testing were almost all within the generally accepted ASHARE guidelines and considered good for carbon dioxide and carbon monoxide. **No actions are recommended at this time regarding carbon dioxide and carbon monoxide IAQ parameters.**

The temperature readings within these ██████████ floors/rooms/areas at times were slightly higher than the generally accepted ASHARE guidelines of 68-76°F for the winter to spring transition season. If there are no complaints from occupants of the areas/rooms/floors being too warm ██████████ recommends no further action is necessary at this time regarding temperature within these areas, however if DFAS, at their discretion, would like to lower the documented temperature in any areas, ██████████ would recommend lowering the temperature a few degrees.

The relative humidity readings within these [REDACTED] floors/rooms/areas were lower than the generally accepted ASHARA relative humidity range of 40-60%, which is considered a proper comfort zone. During the IAQ performed on June 24th through 30th, 2016 the RH readings in these areas were within the ASHARE guidelines. During winter months the RH can decrease easily below the ASHARE guideline "comfort zone". [REDACTED] believes that going into the spring/summer months the RH should increase some to the ASHARE "comfort zone" because of higher humidity outside, a decrease of heating requirements, and the increase in % of outside air. [REDACTED] recommends increasing the relative humidity, if possible, in these areas during the winter months by again increasing the outside air intake however, it would be at the Clients discretion to make any adjustments.

The results from both the airborne ambient particulate (dust) and fibers were lower when compared to the outside. **No actions are recommended at this time regarding ambient particulate (dust) and fibers.**

The very low microbial air results inside the DFAS occupied areas on the [REDACTED] floors, and the sub basement training room within the [REDACTED] building indicted that there was not a source of mold affecting the ambient air. During the site visit on March 1, 2018, there were no visual indicators of suspect mold growth associated with the building materials on the floors, walls or ceiling tiles. [REDACTED] did not observe or detect any suspect mold growth, moisture or odors. **No actions are recommended at this time.**

Enclosed, please find copies of the results of the on-site testing, copies of the laboratory certificates of analysis and chain of custody's, the air monitoring data sheets, IAQ screening general parameters data sheets, and field notes taken on site.

If there are any questions with regard to our report, or if we may be of further service, please contact our office. Thank you for utilizing [REDACTED] as your environmental consulting firm.

[REDACTED]

Arlothue

[REDACTED]

IAQ SCREENING - GENERAL PARAMETERS

Job Site:

DFAS - Select Arms
1240 E. 9th St.
Cleveland, Ohio

Date:

2/27 & 3/1/2018

Job #

18044

Station #	Location	Time	Temp (°F)	RH (%)	CO (ppm)	CO (ppm)	Particulate Location
1	30th fl 30A-5/6 9/10	1026A	72.4	26.1	775	0.0	393
		108P / 453P	75.6	24.7	726	2.1	563
2	28th fl. 283-28 284-48	1032A	74.3	26.0	950	0.2	633
		112P / 456P	75.8	24.8	851	2.1	766
3	285-19	1036A	73.2	25.7	792	0.0	1080
		114P	75.6	24.7	X	X	1020
4	289-16	1044A	75.5	25.5	791	0.1	932
		116P	76.3	24.5	X	X	2400
5	2829B doorway hallway	1048A	76.0	25.2	822	0.1	380
		118P	76.5	24.6	X	X	515
6	26th fl 263-10/12	1053A	75.4	25.1	730	0.2	679
		122P	76.9	24.5	789	1.9	782
7	25th fl 251-43/37	1105A / 1008A	76.2	24.9	893	2.1	796
		125P / 5P	77.1	24.9	695	1.8	2920
8	253-40/39	1109A / 1010A	76.4	24.8	842	1.9	486
		127P	77.4	24.2	X	X	933
9	254-30 27 38	1112A / 1013A	75.8	24.8	837	1.8	783
		128P	77.6	24.2	X	X	981

Sampling Conducted By:

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IAQ SCREENING - GENERAL PARAMETERS

Job Site:

DFAS - Select Arms
1240 E. 9th St.
Cleveland, Ohio

Date:

2/27 & 3/1/2018

Job #

18044

Station	Location	Time	Temp (°F)	RH (%)	CO (ppm)	CO (ppm)	Particulate Location
10	25th fl - 2567D E	1116A 1014A	75.4	25.1	802	1.7	849
		130P	76.9	24.3	X	X	1600
11	24th fl 244-04	1130A 1016A	72.1	26.0	794	1.6	700
		135P 503P	76.2	24.6	655	1.6	953
12	240-25/33	1132A 1017A	73.9	25.7	863	1.7	688
		137P	76.8	24.5	X	X	1080
13	247-42/22 (243)	1134A 1018A	75.1	25.0	780	1.7	957
		139P	76.5	24.3	X	X	1520
14	241-22	1137A 1019A	75.8	25.0	849	1.6	729
		141P	76.4	24.5	X	X	1040
15	19th fl - 1923A downway hallway	1146A 1023A	76.1	25.1	798	1.6	330
		146P 507P	75.8	24.8	690	1.6	399
16	19A-16/40	1149A 1024A	76.4	24.6	851	0.8	4400
		148P	76.2	24.7	X	X	2390
17	196-27 194-47/48	1203P 1026A	76.6	24.6	800	1.5	755
		150P	76.2	24.5	X	X	912
18	9th fl - 9B-03/04 hallway	1213P 1030A	76.4	24.5	695	1.5	671
		154P 510P	76.6	24.4	620	1.5	942

Sampling Conducted By:

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IAQ SCREENING - GENERAL PARAMETERS

Job Site:

DFAS - Select Arms
1240 E. 9th St.
Cleveland, Ohio

Date:

2/21 ~~3/1~~ 2018

Job #

18044

Station	Location	Time	Temp (°F)	RH (%)	CO ₂ (ppm)	CO (ppm)	Particulate Location
19	3rd fl - 03-33 open space	1218P 1033A	75.8	24.7	709	1.5	1400
		158P 512P	77.1	24.2	610	1.5	6630
20	OUTSIDE W. entrance	1010A 1000A	55.4	28.9	375	0.5	7320
		216P 448P	59.5	25.5	407 381	1.7 2.3	11,700
21	SB training room	1254P	73.9	25.6	(L)	(L)	1790
		211P	74.0	25.1	(L)	(L)	6000 ?
X							* New screen & alcohol - didn't change reading
X							
X							
X							
X							
X							

Sampling Conducted By:

[Redacted]

Page

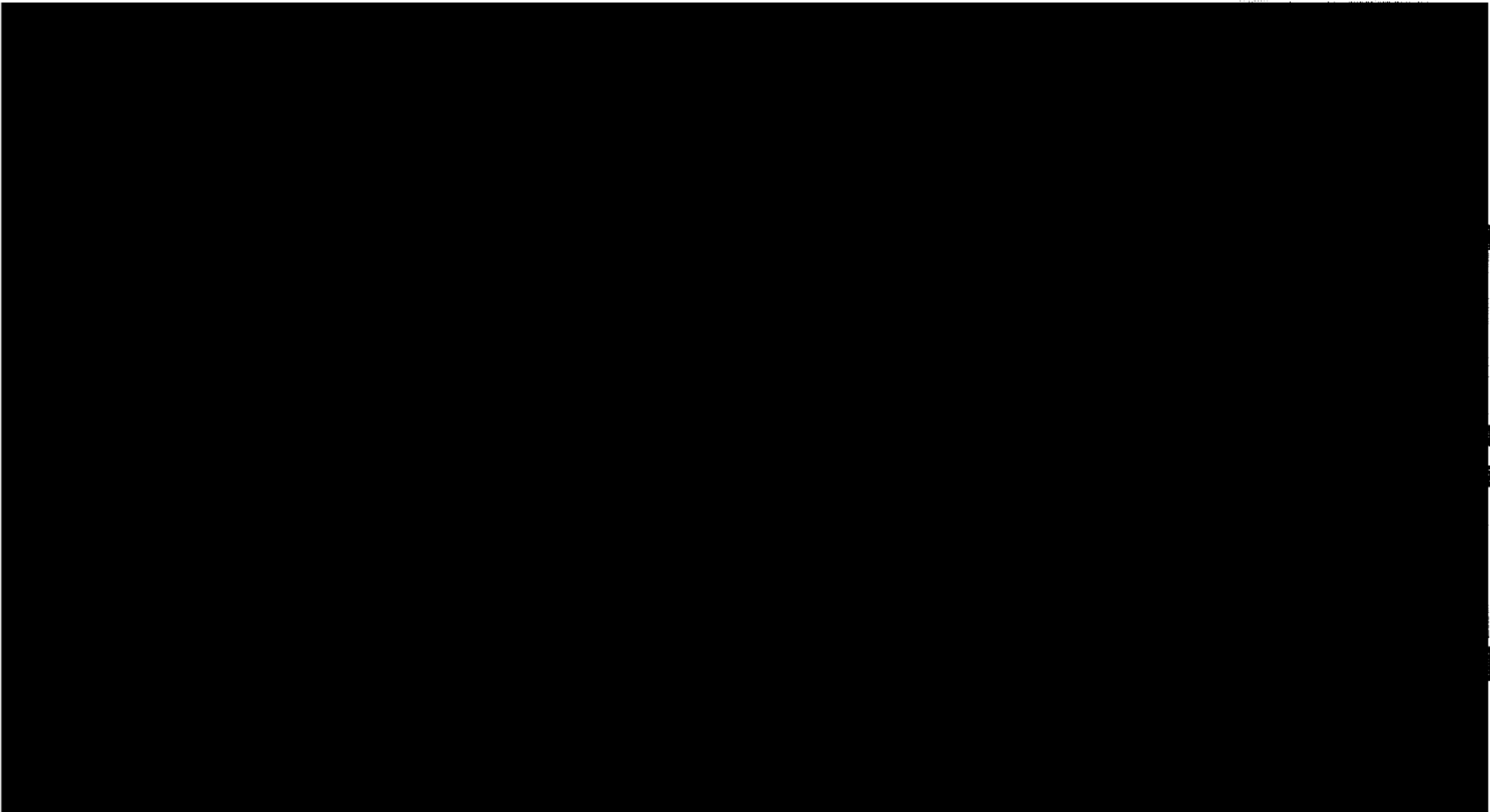
3 of 3

(L) LOCKED

SanAir ID Number	18009189-005			18009189-006			18009189-007			18009189-008		
	105C			105C			105C			105C		
	044 3118 40						044 3118 42			044 3118 43		
	25th - Inside 252-37						24th - 2449 Entry - Adj. To 24A - 26 / 28			24th - Column Adjacent West Of 242-38		
							Air Cassette - Allergenco-D					
Analytical Sensitivity	11						90 Liters					
Background Density				2+			11 Count/M ³			2+		
Other	Raw Count	Count/M ³	%	Raw Count	Count/M ³	%	Raw Count	Count/M ³	%	Raw Count	Count/M ³	%
Mycelial Fragments	1	11	n/a									
Fungal Identification	Raw Count	Count/M ³	%	Raw Count	Count/M ³	%	Raw Count	Count/M ³	%	Raw Count	Count/M ³	%
	1	11	50				1	11	50	1	11	
	1	11	50					11		2	22	
										1		
	2	22		4	44							

Name: CTG Environmental, LLC

SanAir ID Number	18009189-009			18009189-010			18009189-011			18009189-012		
	105C 044 3118 44 19th - Column / 19B-44 Air Cassette - Allergenco-D 90 Liters 11 Count/M³ 2+			105C 								



Organism Descriptions

The descriptions of the organisms presented are derived from various reference materials. The laboratory report is based on the data derived from the samples submitted and no interpretation of the data, as to potential, or actual, health effects resulting from exposure to the numbers of organisms found, can be made by laboratory personnel. Any interpretation of the potential health effects of the presence of this organism must be made by qualified professional personnel with first hand knowledge of the sample site, and the problems associated with that site.

Mycelial Fragments - A mycelium (plural = mycelia) is the "body" of a fungus. It is a collective term for hyphae (singular = hypha), which are the tubular units of the mycelium usually composed of chitin. The terms hyphae and mycelial fragments are used interchangeably. [This information was referenced from the mycology text "The Fifth Kingdom"] In some cases a fungal identification cannot be obtained due to lack of sporulation. Only the mycelial fragments are present, and cannot be identified without the distinguishing characteristics of the spores or the structures they grow from.

Health Effects: Allergic reactions may occur in the presence of spores (conidia) or mycelial/hyphal fragments.

Ascospores - From the fungal Subphylum Ascomycotina. Ascospores are ubiquitous in nature and are commonly found in the outdoor environment. This class contains the "sac fungi" and yeasts. Some ascospores can be identified by spore morphology, however; some care should be exercised with regard to specific identification. They are identified on tape lifts and non-viable analysis by the fact that they have no attachment scars and are sometimes enclosed in sheaths with or without sacs.

Ascomycetes may develop both sexual and asexual stages. Rain and high humidity may help asci to release, and disperse ascospores, which is why during these weather conditions there is a great increase in counts.

Health Effects: This group contains possible allergens.

Aspergillus/Penicillium - These spores are easily aerosolized. Only through the visualization of reproductive structures can the genera be distinguished. Also included in this group are the spores of the genera Acremonium, Phialophora, Verticillium, Paecilomyces, etc. Small, round spores of this group lack the necessary distinguishing characteristics when seen on non-viable examination.

Health Effects: Can cause a variety of symptoms including allergic reactions. Most symptoms occur if the individual is immunocompromised in some way (HIV, cancer, etc). Both Penicillium and Aspergillus spores share similar morphology on non-viable analysis and therefore are lumped together into the same group.

Basidiospores - From the Subphylum Basidiomycotina which contains the mushrooms, shelf fungi, and a variety of other macrofungi. They are saprophytes, ectomycorrhizal fungi or agents of wood rot, which may destroy the structure wood of buildings. It is extremely difficult to identify a specific genera of mushrooms by using standard culture plate techniques. Some basidiomycete spores can be identified by spore morphology; however, some care should be exercised with regard to specific identification. The release of basidiospores is dependant upon moisture, and they are dispersed by wind.

Health Effects: Many have the potential to produce a variety of toxins. Members of this group may trigger Type I and III fungal hypersensitivity reactions. Rarely reported as opportunistic pathogens.

Chaetomium species - It is an ascomycete. It is found on a variety of substrates containing cellulose including paper and plant compost. It can be found on the damp or water damaged paper in sheetrock after a long term water damage. Several species have been reported to play a major role in decomposition of cellulose made materials. These fungi are able to dissolve the cellulose fibers in cotton and paper, and thus cause these materials to disintegrate. The process is especially rapid under moist conditions.

Health Effects: Chaetomium can produce type I fungal hypersensitivity and has caused onychomycosis (nail infections).

References: Flannigan, Brian, Robert A. Samson, and J. David Miller, eds. Microorganisms in Home and Indoor Work Environments: Diversity, Health Impacts, Investigation, and Control. London and New York: Taylor & Francis, 2001.

Organism Descriptions

The descriptions of the organisms presented are derived from various reference materials. The laboratory report is based on the data derived from the samples submitted and no interpretation of the data, as to potential, or actual, health effects resulting from exposure to the numbers of organisms found, can be made by laboratory personnel. Any interpretation of the potential health effects of the presence of this organism must be made by qualified professional personnel with first hand knowledge of the sample site, and the problems associated with that site.

Cladosporium species - The most commonly identified outdoor fungus. The outdoor numbers are reduced in the winter and are often high in the summer. Often found indoors in numbers less than outdoor numbers. It is commonly found on the surface of fiberglass duct liner in the interior of supply ducts. A wide variety of plants are food sources for this fungus. It is found on dead plants, woody plants, food, straw, soil, paint and textiles. Often found in dirty refrigerators and especially in reservoirs where condensation is collected, on moist window frames it can easily be seen covering the whole painted area with a velvety olive green layer.

Health Effects: It is a common allergen. It can cause mycosis. Common cause of extrinsic asthma (immediate-type hypersensitivity: type I). Acute symptoms include edema and bronchiospasms, chronic cases may develop pulmonary emphysema. Illnesses caused by this genus can include phaeohyphomycosis, chromoblastomycosis, hay fever and common allergies.

References: Flannigan, Brian, Robert A. Samson, and J. David Miller, eds. *Microorganisms in Home and Indoor Work Environments: Diversity, Health Impacts, Investigation, and Control*. London and New York: Taylor & Francis, 2001.

Epicoccum species - It is found in plants, soil, grains, textiles, and paper products. Frequently isolated from air and occasionally occurs in house dust. Is a saprophyte and considered a weakly parasitic secondary invader of plants, moldy paper and textiles. Epicoccum is usually isolated with either Cladosporium species or Aureobasidium species.

Health Effects: A common allergen. It also has the potential to produce type I fungal hypersensitivity reactions.

References: Flannigan, Brian, Robert A. Samson, and J. David Miller, eds. *Microorganisms in Home and Indoor Work Environments: Diversity, Health Impacts, Investigation, and Control*. London and New York: Taylor & Francis, 2001.

Smuts/Myxomycetes - Smuts and Myxomycetes are parasitic plant pathogens. They are typically grouped together due to their association with plants, the outdoors and because they share similar microscopic morphology.

Health Effects: Can produce type I fungal hypersensitivity reactions.

References: Martin, G.W., C.J. Alexopoulos, and M.L. Farr. *The Genera of Myxomycetes*. Iowa City, Iowa: University of Iowa Press, 1983.

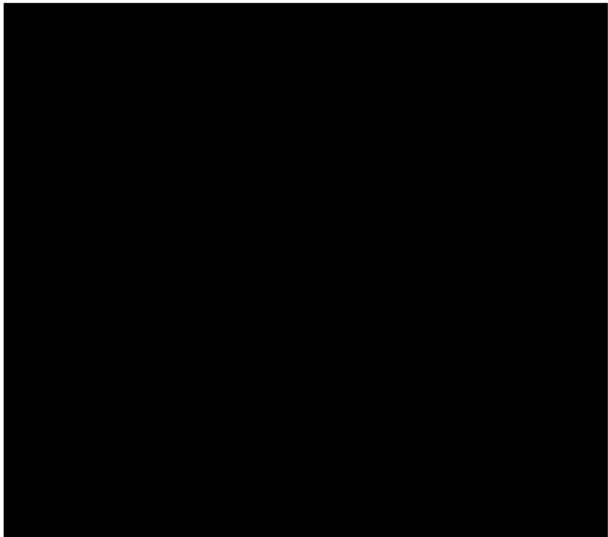


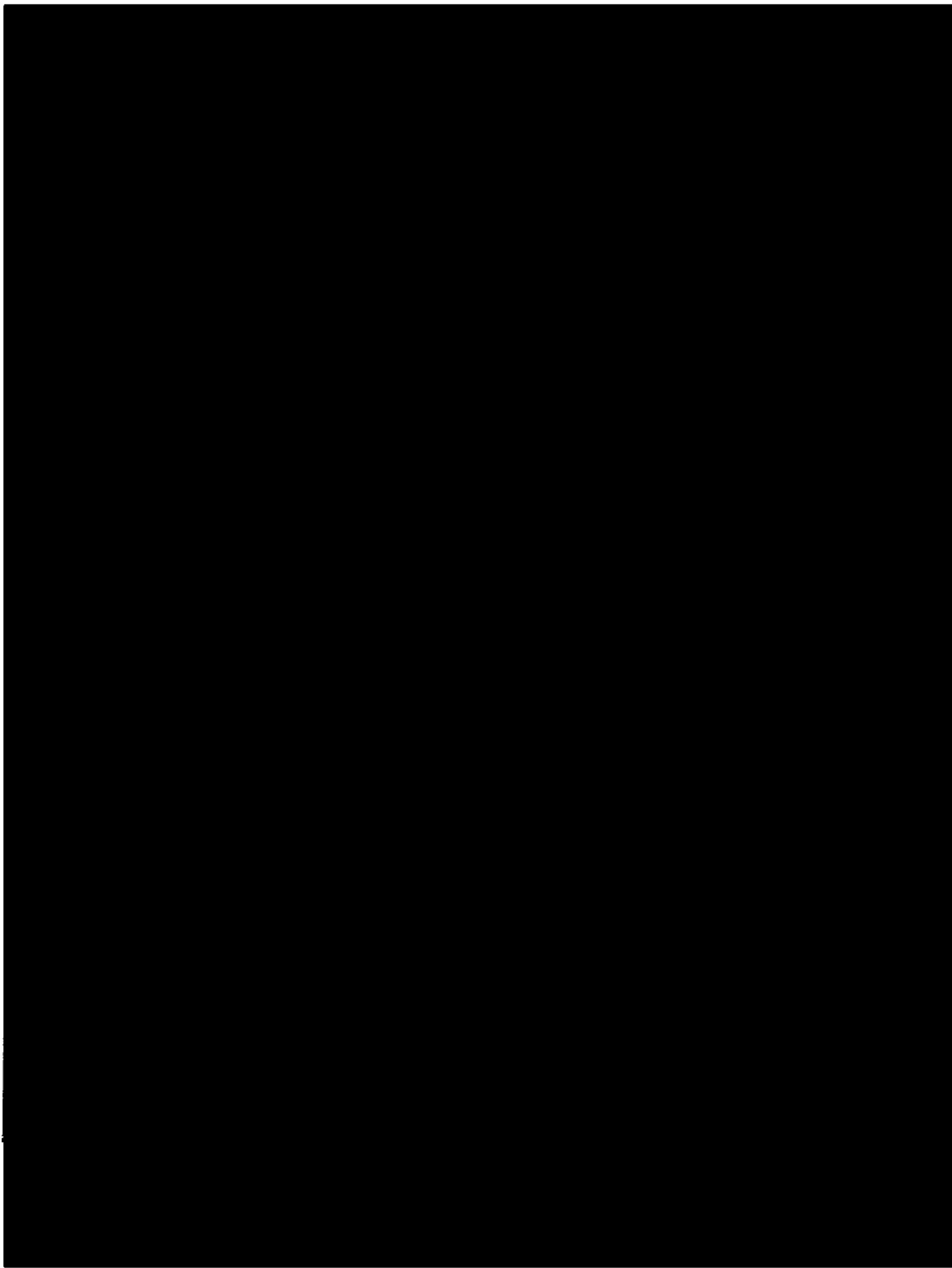
Dear Dave Meyer,

We at SanAir would like to thank you for the work you recently submitted. The 15 sample(s) were received on Monday, March 05, 2018 via FedEx. The final report(s) is enclosed for the following sample(s): 044 3118 49, 044 3118 50, 044 3118 36, 044 3118 37, 044 3118 38, 044 3118 39, 044 3118 40, 044 3118 41, 044 3118 42, 044 3118 43, 044 3118 44, 044 3118 45, 044 3118 46, 044 3118 47, 044 3118 48.

These results only pertain to this job and should not be used in the interpretation of any other job. This report is only complete in its entirety. Refer to the listing below of the pages included in a complete final report.

Sincerely,





PROJECT: DFAS - select floor
1240 E. 9th St.
CLEVELAND, OH

CONTRACTOR: _____

DATE: _____

ACTIVITY: 1AQ+FIBER+MICROBIAL

CTG JOB #: 18044

SAMPLED FOR: METHOD CASSETTE FILTER

☐ Asbestos ☐ 7400 ☐ TEM ☐ 25mm ☐ 37mm ☐ MCE ☐ PC

☐ Dust _____

☐ Gas _____

☐ Vapor _____

☒ Other Microbial

☐ Analytical Method _____ ☐ Ambient Temp. _____

Protection Worn During Sampling:

☐ Full Body Coveralls _____

☐ Respirator-Air Purifying NDN4

☐ Respirator-Power Air Purifying _____

☐ Respirator-Supplied Air _____

PERSONAL AIR SAMPLING

Employee Sampled N/A

Social Security No. _____

Smoker: ☐ Yes ☐ No

Job Function _____

Description of Activity: _____

SAMPLE NO.	DATE	SAMPLE LOCATION
044 3118 36	3/1/18	30 th - 30A-17/18
044 3118 37	↓	28 th - Column between 281-20 & 281-8
044 3118 38		- 285-47
044 3118 39		26 th - 2669D dm threshold, facing out to cubicles
044 3118 40		25 th - inside 252-37
044 3118 41		- 25A/7-8 ext. corner w/ hallway
044 3118 42	↓	24 th - 2449 entry - adj. to 24A-26/28

SAMPLE NO.	SAMPLING TIMES			FLOW RATES		INSTRUMENT SERIAL NO.	SAMPLE TYPE				SAMPLE LOCATION		
	START	STOP	TOTAL (MIN.)	START	STOP (LITERS/MIN.)		AMBIENT	AREA	PERSONAL	FINAL	IN AREA	OUTSIDE AREA	VOL.
044 3118-36	4:12 P	4:18 P	6	15.0	15.0	25757	✓				✓		90
-37	4:02 P	4:08 P	6	15.0	15.0		✓				✓		90
-38	3:52 P	3:58 P	6	15.0	15.0		✓				✓		90
-39	3:21 P	3:27 P	6	15.0	15.0		✓				✓		90
-40	3:07 P	3:13 P	6	15.0	15.0		✓				✓		90
-41	2:58 P	3:04 P	6	15.0	15.0		✓				✓		90
✓-42	12:40 P	12:46 P	6	15.0	15.0	✓	✓				✓		90

Inspector: _____

AIR MONITORING DATA SHEET

PROJECT:		SAMPLE NO.	DATE	SAMPLE LOCATION
DFAS - select & remove 1240 E. 9th St. CLEVELAND, OH		044 3118 43	3/1/18	24th - column adjacent ↓ West of 242-B8
CONTRACTOR: _____ DATE: _____ ACTIVITY: IAQ + FIBER + MICROBIAL CTG JOB #: 18044		044 3118 44		19th - COLUMN / 19B-44 ↓
SAMPLED FOR: METHOD CASSETTE FILTER ☐ Asbestos ☐ 7400 ☐ TBA ☐ 25mm ☐ 37mm ☐ MCE ☐ FO ☐ Dust _____ ☐ Gas _____ ☐ Vapor _____ ☒ Other Microbial ☐ Analytical Method _____ ☐ Ambient Temp. _____		044 3118 45		↓ - 195-32 outside
Protection Worn During Sampling: ☐ Full Body Coveralls NONE ☐ Respirator-Air Purifying _____ ☐ Respirator-Power Air Purifying _____ ☐ Respirator-Supplied Air _____		044 3118 46		9th - <u>9B-15/16</u> <u>9B-39/40</u>
PERSONAL AIR SAMPLING Employee Sampled N/A Social Security No. _____ Smoker: ☐ Yes ☐ No Job Function _____ Description of Activity: _____		044 3118 47		3rd - Column - Adj. E. of 3B-23/24
		044 3118 48		SB - NW corner of Training ROOM (SB39)
		044 3118 49		OUTSIDE - W. entrance, N. outlet on deck

SAMPLE NO.	SAMPLING TIMES			FLOW RATES		INSTRUMENT SERIAL NO.	SAMPLE TYPE				SAMPLE LOCATION		
	START	STOP	TOTAL (MIN.)	START	STOP (LITERS/MIN.)		AMBIENT	AREA	PERSONAL	FINAL	IN AREA	OUTSIDE AREA	VOL.
44													
18-43	12:31p	12:37p	6	15.0	15.0	25757	✓				✓		90
1-44	11:56a	12:02p	6	15.0	15.0		✓				✓		90
-45	12:10p	12:12p	6	15.0	15.0		✓				✓		90
-46	2:22p	2:28p	6	15.0	15.0		✓				✓		90
-47	2:11p	2:17p	6	15.0	15.0		✓				✓		90
-48	1:58p	2:04p	6	15.0	15.0		✓				✓		90
✓-49	4:28p	4:34p	6	15.0	15.0	✓	■					■	■

Client: DFAS - Cleveland
 Location: Select gl mr
 Contractor: [REDACTED]
 Industrial Hygienist: [REDACTED]

Job # 18044
 Date: 2/27 & 3/1/2018
 Total # of Samples: 13 + B = 14
 Signature: [REDACTED]

Sample Number	Sample Type	Sample Location	Start Time / Flow	Stop Time / Flow	Total Time (min)	Volume (liters)	Fibers/ fields	F/ cc	Pump ID	Site Activity
044 22718 22	AM I	30 th fl. - 30A-03/04 30B-39/40	245 pm 10.0	345 pm 10.0	60	600	0 / 100	<0.005	25757	
044 22718 23	AM I	28 th fl. - 289-20	250 pm 10.0	350 pm 10.0	60	600	0 / 100	<0.005	22369	
044 22718 24	AM I	↓ Column - 2 nd N of S, 2 nd E of W	256 pm 10.0	356 pm 10.0	60	600	0.5 / 100	<0.005	24652	
044 22718 25	AM I	26 th fl. - Column - 2 nd S of N, 2 nd E of W	303 pm 10.0	403 pm 10.0	60	600	0.5 / 100	<0.005	22338	
044 3118 26	AM I	25 th fl. - Column adj. S. of cub. 71 25323	932 am 10.0	942 am 10.0	70	700	2 / 100	<0.005 ⁴	24652	
044 3118 27	AM I	↓ Column - 2 nd col w. of E, 5 th N	936 am 10.0	940 am 10.0	64	640	0 / 100	<0.005 ⁴	22369	
044 3118 28	AM I	24 th fl. - 244-36/08 ↓	943 am 10.0	247 am 10.0	64	640	3.5 / 100	<0.005 ⁴	22338	

KEY:

AS- AREA SAMPLE
 PS- PERSONAL SAMPLE
 FC- FINAL CLEARANCE
 AM- AMBIENT SAMPLE

I- INSIDE WORK AREA
 O- OUTSIDE WORK AREA
 B- BLANK

COMMENTS

LOT 20170116
 PCM 0.8 MCE 25mm

Client: DFAS - Cleveland
 Location: Select gl mr
 Contractor: _____
 Industrial Hygienist: _____

Job # 18044
 Date: 2/27 & 3/1/2018
 Total # of Samples 13+B=14
 Signature: _____

Sample Number	Sample Type	Sample Location	Start Time / Flow	Stop Time / Flow	Total Time (min)	Volume (liters)	Fibers/ fields	F/ cc	Pump ID	Site Activity
044 3/18 29	AM I	24th - refrigerator by 249-04	1104 AM 10.0	1124 PM 10.0	80	800	0.5 100	<0.003	24652	
044 3/18 30	AM I	19th - CER cabinet 196-24 (ave)	1113 AM 10.0	1213 PM 10.0	60	600	3 100	<0.005	22338	
044 3/18 31	AM I	- COLUMN 19D-30 area	1118 AM 10.0	1218 PM 10.0	60	600	2.5 100	<0.005	22369	
044 3/18 32	AM I	9th - printing 9A-37/38	1:50 PM 10.0	2:50 PM 10.0	60	600	1 100	<0.005	22338	
044 3/18 33	AM I	3rd - B41/42	1:42 PM 10.0	2:12 PM 10.0	Someone unplugged				22369	Restart 2:39pm Stopped 3:39pm
044 3/18 34	AM II	Subbasement (SB39) Training - pm (NE) - corner duct run opposite office	1:34 PM 10.0	2:34 PM 10.0	60	600	1 100	<0.005	24652	Moving furniture to run cables next day
044 3/18 35	- B	BLANK	X 10.0	X 10.0			0 100			

KEY:
 AS- AREA SAMPLE
 PS- PERSONAL SAMPLE
 FC- FINAL CLEARANCE
 AM- AMBIENT SAMPLE

I- INSIDE WORK AREA
 O- OUTSIDE WORK AREA
 B- BLANK

COMMENTS
Same @ pg 1

Sampling Data

BLI Project#: R102515
Project Name: CTG ENVIRONMENTAL-18044 DFAS AJC 9 FLRS

Date Sampled: 2/27/18
Sampled By: CLIENT
Date Analyzed: 3/12/18

Analytical Parameters

Effective Filter Area (mm²): 385

Graticule Field Area (mm²): 0.00785

Lab Sample #	Client Sample #	Sample Location	Sample Type	Sample Volume(L)	Fields	Fibers	Detection Limit(F/cc)	Results	
								F/mm ²	F/cc
978187	04422718-22	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A
978188	04422718-23	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A
978189	04422718-24	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A
978190	04422718-25	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A
978191	F443118-26	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A

ANALYST:

REVIEWED BY:

N.C. Batta / R. Shumate
(QA/QC Officer)

- * Results pertain only to the items tested.
- * This report does not constitute endorsement by AIHA and/or any other U.S. government agencies.
- * Named analyst may not be sole analyst. Refer to Chain of Custody for additional analysts.
- * Sample volumes are calculated from data supplied by the client. Batta Laboratories, Inc. does not accept liability for results expressed in fibers per cubic centimeter. Furthermore, Batta Laboratories assumes no responsibility for the accuracy of results reflected by the use of improper collection techniques or equipment.
- * Current YTD Sr value is 0.33 for intralaboratory and 0.28 for interlaboratory. This value may change slightly over ti
- * Sample results listed above are not blank-corrected. NIOSH 7400 requires submission of blanks (minimum 2 or 10%) with samples. Batta assumes no responsibility for collection inconsistent with the method.
- * MU Values based on inter-lab data: 5-20 f/100 fields: (LCL: 13.6, UCL: 16.9), 20.5-50 f/100 fields: (LCL: 39.9, UCL 46.1), >50 f/100 fields: (LCL: 137.8, UCL: 176.0)

CERTIFICATE OF PCM ANALYSIS

Page 2 of 3

Test Method: NIOSH 7400, Issue 2: 15 August 1994

Report Date:

3/12/2018

Sampling Data

BLI Project#: R102515
Project Name: CTG ENVIRONMENTAL-18044 DFAS AJC 9 FLRS

Date Sampled: 3/1/18
Sampled By: CLIENT
Date Analyzed: 3/12/18

Analytical Parameters

Effective Filter Area (mm²): 385Graticule Field Area(mm²): 0.00785

Lab Sample #	Client Sample #	Sample Location	Sample Type	Sample Volume(L)	Fields	Fibers	Detection Limit(F/cc)	Results	
								F/mm ²	F/cc
978192	0443118-27	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A
978193	0443118-28	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A
978194	0443118-29	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A
978195	0443118-30	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A
978196	0443118-31	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A

ANALYST:

REVIEWED BY:

(QA/QC Officer)

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- * MU Values based on inter-lab data: 5-20 f/100 fields: (LCL: 13.6, UCL: 16.9), 20.5-50 f/100 fields: (LCL: 39.9, UCL: 46.1), >50 f/100 fields: (LCL: 137.8, UCL: 176.0)

Sampling Data

BLI Project#: R102515
Project Name: CTG ENVIRONMENTAL-18044 DFAS AJC 9 FLRS

Date Sampled: 3/1/18
Sampled By: CLIENT
Date Analyzed: 3/12/18

Analytical Parameters

Effective Filter Area (mm²): 385

Graticule Field Area (mm²): 0.00785

Lab Sample #	Client Sample #	Sample Location	Sample Type	Sample Volume(L)	Fields	Fibers	Detection Limit(F/cc)	Results	
								F/mm ²	F/cc
978197	0443118-32	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A
978198	0443118-33	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A
978199	0443118-34	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A
978200	0443118-35	N/A	PCM	0	100	<5.5	N/A	<7.0	N/A

ANALYST: [REDACTED]

- * Results pertain only to the items tested.
- * This report does not constitute endorsement by AIHA and/or any other U.S. government agencies.
- * Named analyst may not be sole analyst. Refer to Chain of Custody for additional analysts.
- * Sample volumes are calculated from data supplied by the client. Batta Laboratories, Inc. does not accept liability for results expressed in fibers per cubic centimeter. Furthermore, Batta Laboratories assumes no responsibility for the accuracy of results reflected by the use of improper collection techniques or equipment.
- * Current YTD Sr value is 0.33 for intralaboratory and 0.28 for interlaboratory. This value may change slightly over time.
- * Sample results listed above are not blank-corrected. NIOSH 7400 requires submission of blanks (minimum 2 or 10%) with samples. Batta assumes no responsibility for collection inconsistent with the method.
- * MU Values based on inter-lab data: 5-20 f/100 fields: (LCL: 13.6, UCL: 16.9), 20.5-50 f/100 fields: (LCL: 39.9, UCL 46.1), >50 f/100 fields: (LCL: 137.8, UCL: 176.0)

Tel 1: 216 661-6696

Tel 2: _____

Shipping Information

☐ Picked up by BATTA

☐ Delivered by Customer

☒ Shipped by Customer

☒ 3 Hours/Immediate (Note 1) ☐ 72 Hours (Note 5)
☒ 6 Hours/Same Day (Note 2) ☒ 5 Days (Note 6)
☒ 24 Hours (Note 3) ☐ 5 - 10 days
☒ 48 Hours (Note 4) ☐ Other _____

Client Project #: 18044

RUBSIS

4 Unless a specific time is requested, results are guaranteed by 5 p.m. on the 2nd business day.

Project Name: PFAS IAQ select PARA

Project Location: ATC-9 floor

[illegible]

Date: 11/11/2011 Time: 11:11 AM

ONLY ANALYZE MATERIAL LISTED ON BAG OR SAMPLE DATA SHEET

Logged-In By: VM Log-In Date: 3/5/18 Date: 1/10/18

ISSUE ONLY Press any single key to advance to the next screen. Press any single key to return to the menu below.		Method of payment ☐ Cash ☒ Visa/Mastercard/Discover ☐ Money Order ☐ Gift Certificate ☐ Credit ☐ Debit ☐ American Express ☐ Company Card ☐ Other	
Make change ☐ No change please			
Accounting Office Use Only			

For drinking water samples: for results to be valid, lab must receive samples on ice and within 48 hours of collection. For air samples collected by NIOSH 7400 and 7402: in accordance with these NIOSH methods, two field blanks (or 10% of the number of field samples submitted, whichever is greater) must be submitted and be analyzed with field samples.

DATE:

2/27 - 3/1/2018

JOB NUMBER:

18044

GENERAL OBSERVATIONS

04422718-22	078187	0/100	
↓ -23	188	0/100	
-24	189	0.5/100	
↓ -25	190	0.5/100	
0443118-26	191	0.5/100	
-27	192	2/100	
-28	193	0/100	(14) Sampler
-29	194	3.5/100	
-30	195	0.5/100	
-31	196	3/100	
-32	197	2.5/100	
-33	198	1/100	
↓ -34	199	1/100	
-35	200	0/100	

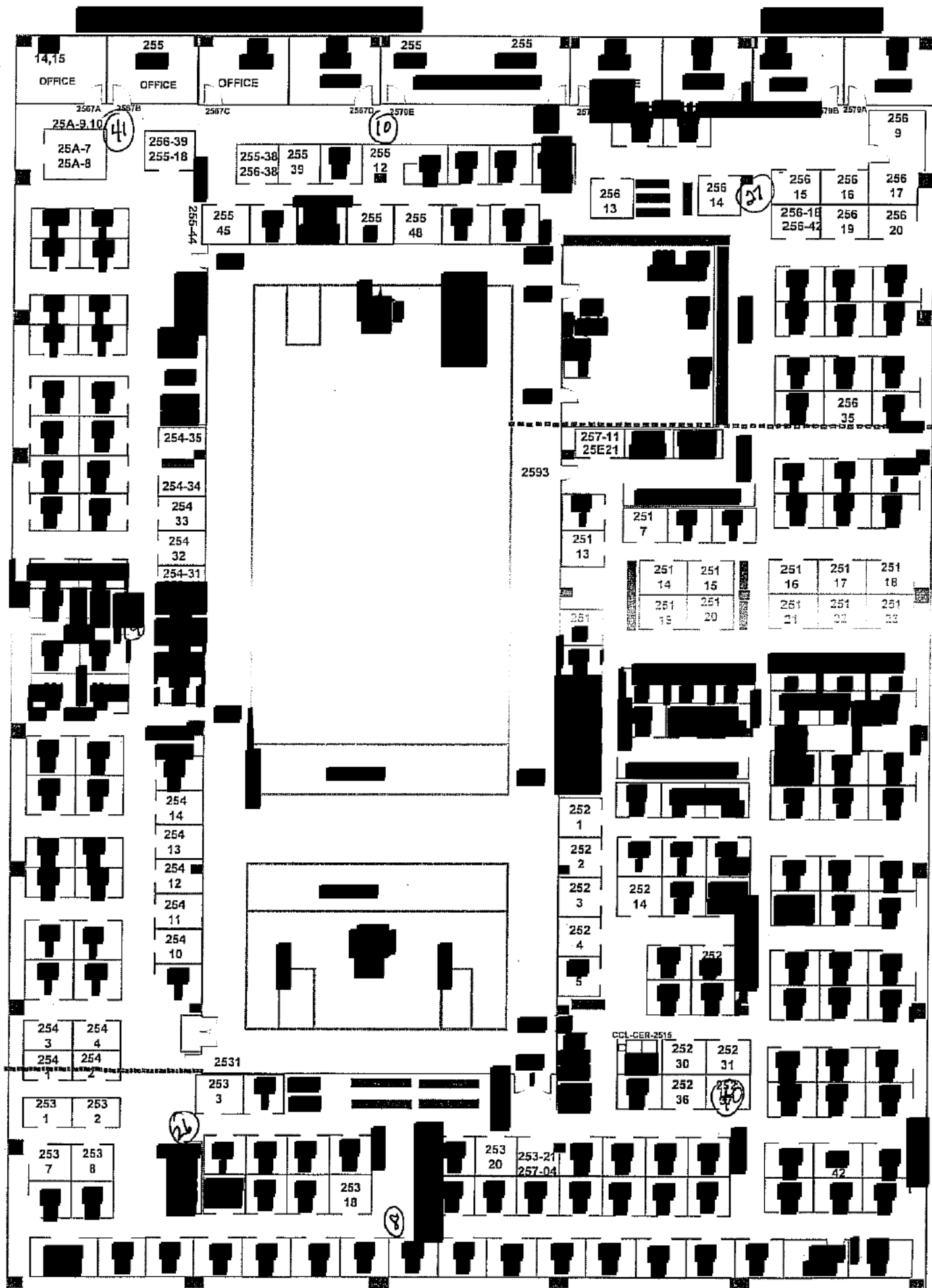
N
↑

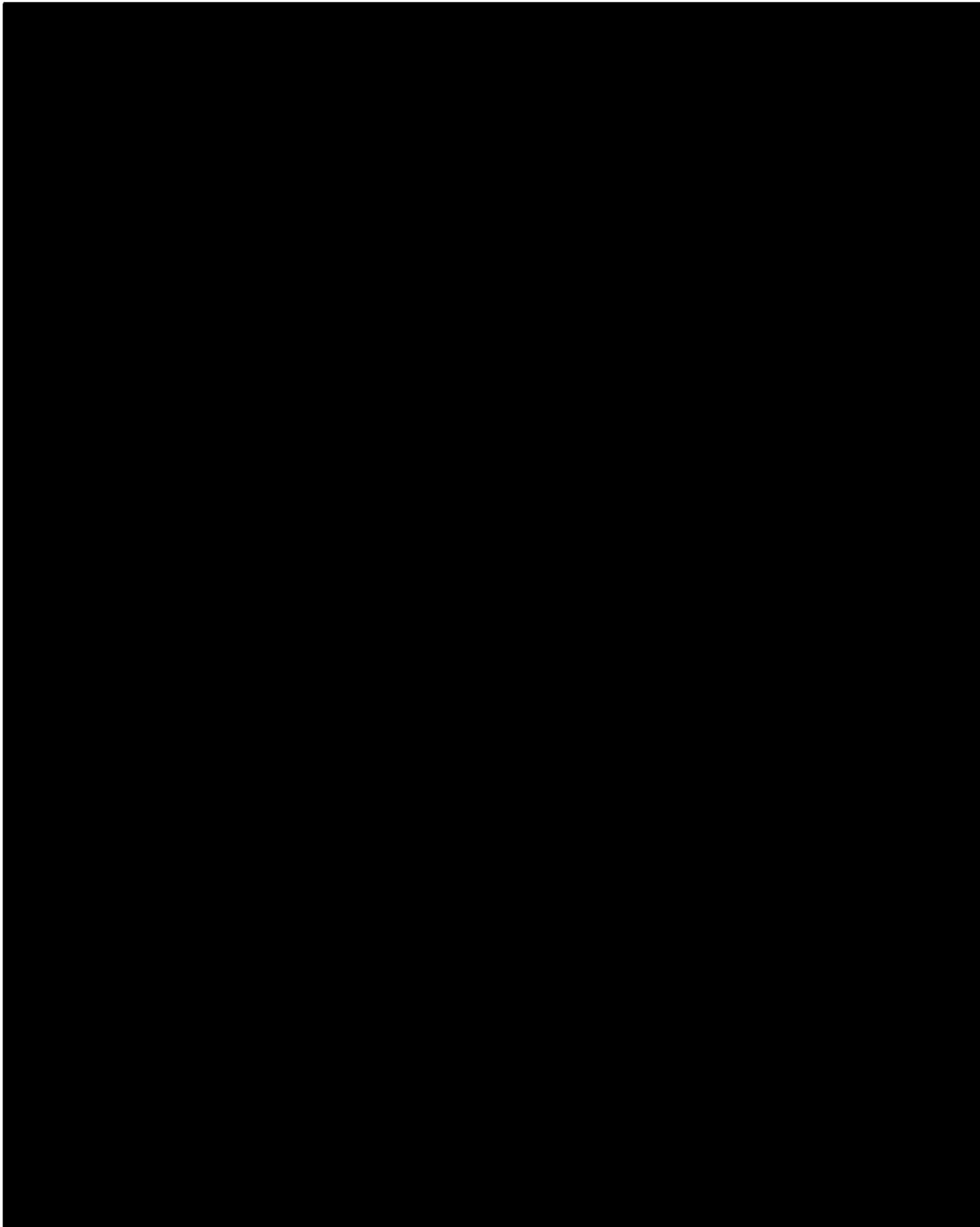
Handwritten scribbles

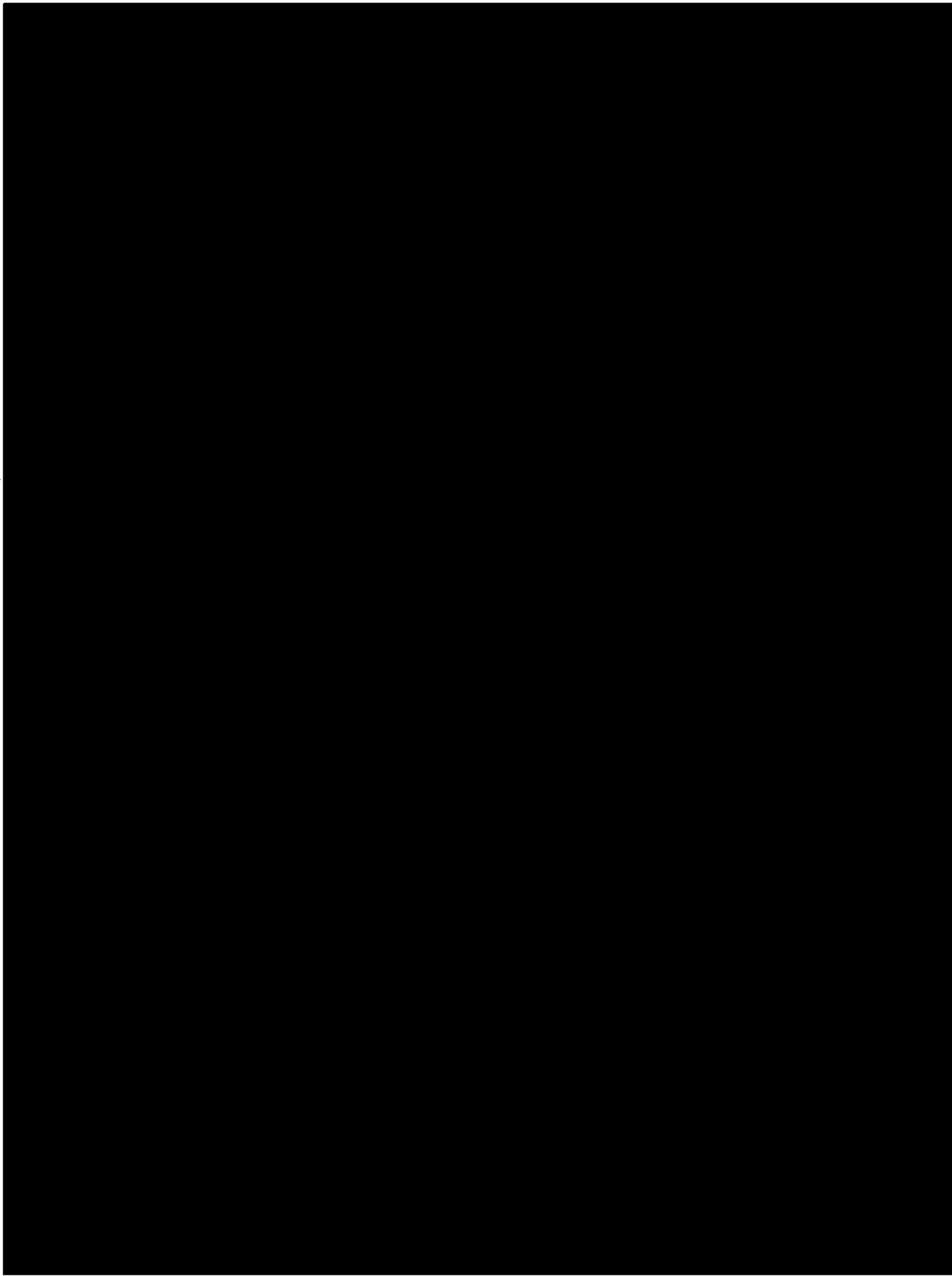


NA

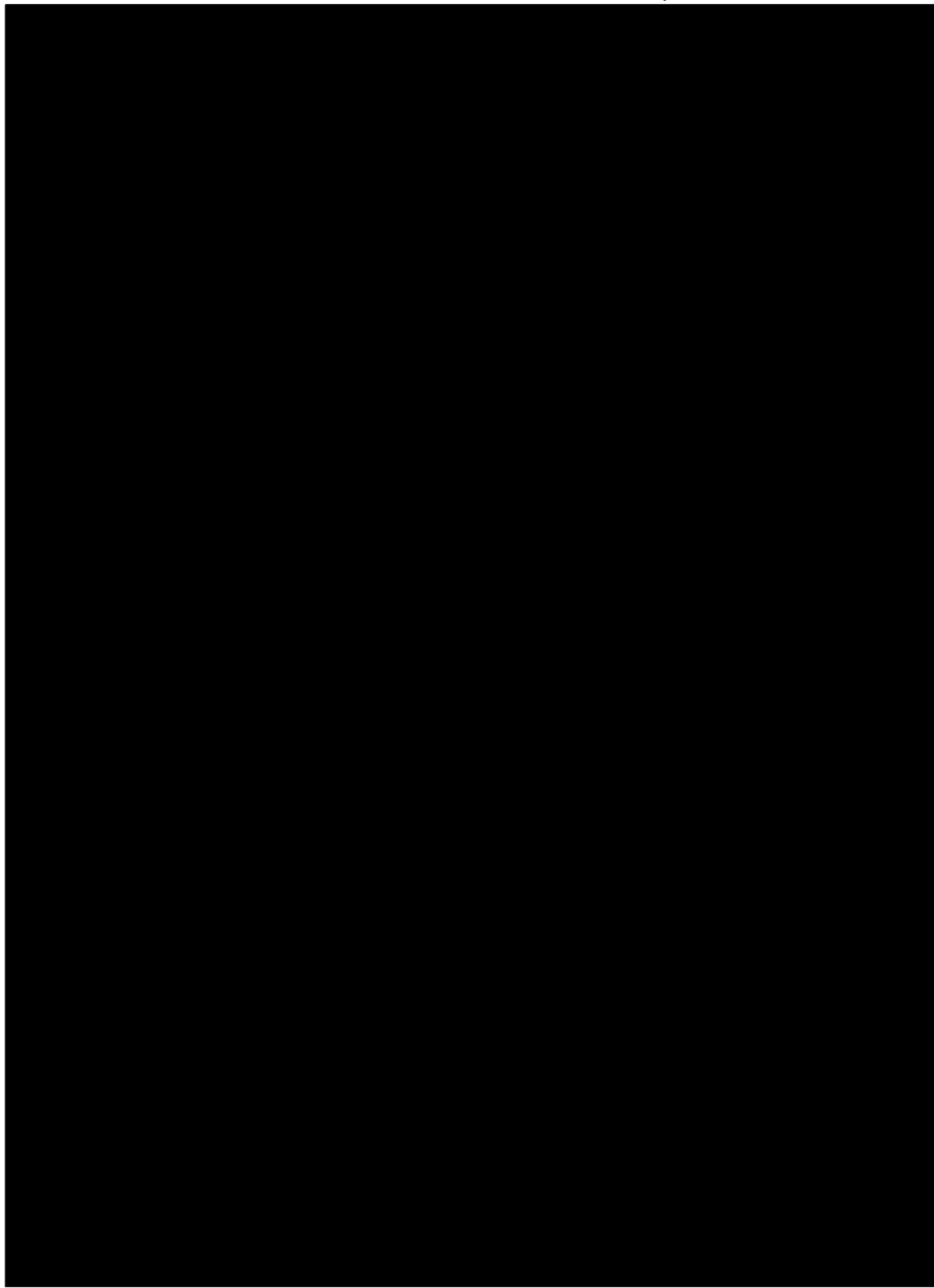
CP











Appendix A - AICFB

(PARTICULATE)

W/
(SAME H)

FLOORS	FOOTAGE	APPROX. DFAS OCCUPATION	MICROBIAL AIR SAMPLING	AIRBORNE AMBIENT DUST SAMPLING	AIRBORNE FIBERS	INITIAL AIR QUALITY EVALUATION	Initial IAQ Review/Investigation	Walk-through Survey	Total Samples
3	8,078	25%	X	X	X	X	1	1	6
9	4,700	15%	X	X	X	X	1	1	6
14	26,541	100%	2	4	2	1	1	1	11
15	27,146	100%	2	4	2	1	1	1	11
16	3,532	15%	1	1	1	1	1	1	6
17	27,651	100%	2	4	2	1	1	1	11
18	27,881	100%	2	4	2	1	1	1	11
19	22,884	80%	X	X	X	X	1	1	10
22	26,344	100%	2	4	2	1	1	1	11
23	27,521	100%	2	4	2	1	1	1	11
24	27,334	100%	X	X	X	X	1	1	11
25	27,304	100%	X	X	X	X	1	1	11
26	7,326	15%	X	X	X	X	1	1	6
27	28,405	100%	2	4	2	1	1	1	11
28	28,426	100%	X	X	X	X	1	1	11
29	28,656	100%	2	4	2	1	1	1	11
30	4,400	15%	X	X	X	X	1	1	6
SB	12,993	15%	X	X	X	1.0	1	1	6
									0
Total # tests per year = 1			30	53	30	18	18	18	167
Total Blanks			4 (one per visit)		4 (one per visit)				8

OUTSIDE 2/27
3/1 completed

3/1 completed

T/RH only - make Qtrak 7565 2/27
3/1 completed

INSTRUMENT CALIBRATION REPORT

Instrument ID 13927
Description TSI 7565-X Q-Trak
Calibrated 6/26/2017

Manufacturer TSI
Model Number 7565-X
Serial Number 7565X0917018
Location New Jersey
Temp 76

Classification
Status pass
Frequency Yearly EOM
Department Lab
Humidity 32

Calibration Specifications

Group # 1
Group Name Barometric Pressure
Stated Accy Per of Reading

Range Acc % 0.0000
Reading Acc % 3.0000
Plus/Minus 0.000

<u>Nom In Val / In Val</u>	<u>In Type</u>	<u>Out Val</u>	<u>Out Type</u>	<u>End As</u>	<u>Lft As</u>	<u>Dev%</u>	<u>Pass/Fail</u>
30.000 / 29.940	inHg	29.940	inHg	29.930	29.940	0.00%	Pass

Test Instruments Used During the Calibration

<u>Test Instrument ID</u>	<u>Description</u>	<u>Manufacturer</u>	<u>Serial Number</u>	<u>(As Of Cal Entry Date)</u>	
				<u>Last Cal Date</u>	<u>Next Cal Date</u>
OMEGA HX93AC/DP25- E	Omega HX93AC/DP25-E	Omega Engineering	1010368 035025 035026	9/15/2016	9/15/2018
OMEGA PX02K1-16A5T /DP25-E-A	Omega PX02K1-16A5T/DP25-E-A	Omega Engineering	168377/8375030	9/15/2016	9/15/2018
OMEGA WT4401-D	Omega WT4401-D	Omega Engineering	101105	9/15/2016	9/15/2018

Notes about this calibration

Calibration Result Calibration Successful

Who Calibrated

hereby certifies that this instrument is calibrated and functions to meet the manufacture's specifications using NIST traceable standards, or is derived from accepted values of physical constants.

INSTRUMENT CALIBRATION REPORT

Instrument ID 23832
Description TSI 982 Probe
Calibrated 2/27/2018 4:49:10PM

Manufacturer	Tsi	State Certified
Model Number	982	Status Pass
Serial Number/ Lot Number	P13100046	Temp °C 25.1
Location	Florida	Humidity % 38
Department		

Calibration Specifications

Group # 1
Group Name Carbon Dioxide
Stated Accy Pct of Reading

Range Acc % 0.0000
Reading Acc % 3.0000
Plus/Minus 0

<u>Nom In Val / In Val</u>	<u>In Type</u>	<u>Out Val</u>	<u>Out Type</u>	<u>End As</u>	<u>Lft As</u>	<u>Dev%</u>	<u>Pass/Fail</u>
1000 / 1000	PPM	1000	PPM	936	1,000	0.00%	Pass

Group # 2
Group Name Carbon Monoxide
Stated Accy Pct of Reading

Range Acc % 0.0000
Reading Acc % 3.0000
Plus/Minus 0.0

<u>Nom In Val / In Val</u>	<u>In Type</u>	<u>Out Val</u>	<u>Out Type</u>	<u>End As</u>	<u>Lft As</u>	<u>Dev%</u>	<u>Pass/Fail</u>
100.0 / 100.0	PPM	100.0	PPM	99.5	100.0	0.00%	Pass

Test Instruments Used During the Calibration

(As Of Cal Entry Date)

<u>Test Standard ID</u>	<u>Description</u>	<u>Manufacturer</u>	<u>Model Number</u>	<u>Serial Number / Lot Number</u>	<u>Last Cal Date / Opened Date</u>	<u>Next Cal Date / Expiration Date</u>
FL 100 CO -/1000 CO2 HAQ-375-1	100 CO/1000 CO2	Gasco		HAQ-375-1		7/7/2020
FL 99.9999% N2 JAP-114-14	N2 - 99.999% JAP-114-14	Gasco	GP12307	JAP-114-14		10/27/2019

Notes about this calibration

Calibration Result Calibration Successful
Who Calibrated

INSTRUMENT CALIBRATION REPORT

Classification

Status pass

Frequency Yearly EOM

Department Lab

Humidity 33

Calibration Specifications

1

Carbon Dioxide

Stated Accy Pet of Reading

Range Acc % 0.0000

Reading Acc % 3.0000

Plus/Minus 0.00

<u>Nom In Val / In Val</u>	<u>In Type</u>	<u>Out Val</u>	<u>Out Type</u>	<u>End As</u>	<u>Lft As</u>	<u>Dev%</u>	<u>Pass/Fail</u>
0.00 / 0.00	ppm	0.00	ppm	0.00	0.00	0.00%	Pass
1000.00 / 1000.00	ppm	1000.00	ppm	987.00	990.00	-1.00%	Pass

Group # 2

Group Name Carbon Monoxide

Stated Accy Pet of Reading

Range Acc % 0.0000

Reading Acc % 3.0000

Plus/Minus 0.00

<u>Nom In Val / In Val</u>	<u>In Type</u>	<u>Out Val</u>	<u>Out Type</u>	<u>End As</u>	<u>Lft As</u>	<u>Dev%</u>	<u>Pass/Fail</u>
0.00 / 0.00	ppm	0.00	ppm	0.00	0.00	0.00%	Pass
100.00 / 100.00	ppm	100.00	ppm	140.30	99.00	-1.00%	Pass

Group # 3

Group Name Relative Humidity

Stated Accy Pet of Reading

Range Acc % 0.0000

Reading Acc % 3.0000

Plus/Minus 0.00

<u>Nom In Val / In Val</u>	<u>In Type</u>	<u>Out Val</u>	<u>Out Type</u>	<u>End As</u>	<u>Lft As</u>	<u>Dev%</u>	<u>Pass/Fail</u>
30.00 / 39.40	%	39.40	%	44.60	39.50	0.25%	Pass

Group # 4

Group Name Temperature

Stated Accy Plus / Minus

Range Acc % 0.0000

Reading Acc % 0.0000

Plus/Minus 1.00

<u>Nom In Val / In Val</u>	<u>In Type</u>	<u>Out Val</u>	<u>Out Type</u>	<u>End As</u>	<u>Lft As</u>	<u>Dev%</u>	<u>Pass/Fail</u>
70.00 / 75.00	°F	75.00	°F	76.40	75.00	0.00%	Pass

INSTRUMENT CALIBRATION REPORT

Instrument ID 23832

Description TSI 982 Probe

Calibrated 6/27/2017

Test Instruments Used During the Calibration

<u>Test Instrument ID</u>	<u>Description</u>	<u>Manufacturer</u>	<u>Serial Number</u>	<u>(As Of Cal Entry Date)</u>	
				<u>Last Cal Date</u>	<u>Next Cal Date</u>
CO/CO2_34LS-375	100 ppm CO, 1000 ppm CO2	Calgaz	MAO-375-1		6/9/2019
NITROGEN	Nitrogen 99.999%	Liquid Technology	7727-37-9	6/1/2016	6/1/2019
OMEGA HX93AC/DP25-E	Omega HX93AC/DP25-E	Omega Engineering	1010368 035025 035026	9/15/2016	9/15/2018
OMEGA PX02K1-16A5T /DP25-E-A	Omega PX02K1-16A5T/DP25-E-A	Omega Engineering	168377/8375030	9/15/2016	9/15/2018
ZERO_AIR_105 1.-1	Zero Grade Air THC <1.0 PPM	Liquid Technology	KAP-A-10	10/1/2015	10/20/2019

Notes about this calibration

Calibration Result Calibration Successful

Who Calibrated [REDACTED]

[REDACTED] hereby certifies that this instrument is calibrated and functions to meet the manufacture's specifications using NIST traceable standards, or is derived from accepted values of physical constants.

INSTRUMENT CALIBRATION REPORT

Instrument ID 14117
Description TSI P-TRAK 8525
Calibrated 2/23/2018 10:16:08AM

Manufacturer Tsi
Model Number 8525
Serial Number/ Lot 8525-12080017
Number
Location Ohio
Department

Calibration Specifications

Group # 1
Group Name
Test Performed: Yes **As Found Result:** Pass

Test Instruments Used During the Calibration

<u>Test Standard ID</u>	<u>Description</u>	<u>Manufacturer</u>
-------------------------	--------------------	---------------------

Notes about this calibration

Calibration Result Calibration Successful
Who Calibrated



CERTIFICATE OF TESTING

[REDACTED]

CALIBRATION STANDARDS USED
PortaCount Bench 2

MODEL	P-TRAK [®] Ultrafine Particle Counter 8525
SERIAL NO.	8525-12080017

VERIFICATION DATA (PARTICLE CONCENTRATION)

TESTING NUMBER	MEASURED CONCENTRATION IN <i>Particles/cm³</i> Tolerance: 95% to 105% of standard		
	TESTING STANDARD	INSTRUMENT OUTPUT	PERCENT OF STANDARD
1	123.3	121.0	98.1
2	323.7	322.2	99.5
3	723.4	715.9	99.0
4	2279.9	2220.7	97.4
5	6490.2	6588.7	101.5

* Indicates out of tolerance condition

[REDACTED] does hereby certify that the above described instrument conforms to the original manufacturer's specifications (not applicable to As Found data) and has been calibrated using standards whose accuracies are traceable to the National Institute of Standards and Technology within the limitations of NIST's calibration services or have been derived from accepted values of natural physical constants or have been derived by the ratio type of self calibration techniques. [REDACTED] is registered to ISO-9001:2008, Quality Assurance Requirements. This report may not be reproduced, except in full, unless permission for the publication of an approved abstract is obtained in writing from the calibration organization issuing this report.

Measurement Variable	System ID Number	Date Last Calibrated	Calibration Due Date
DC Voltage	E002456	10-21-16	04-30-18
DC Voltage	E002794	10-21-16	04-30-18
Particle Concentration	E002040	07-25-16	07-25-17
Particle Concentration	E002790	07-30-16	07-30-17
Particle Diameter	E005270	07-21-16	07-21-17

Calibration procedure used: 10000013472

Overall Rating: PASS

[REDACTED]

Calibrated By

Apr. 28, 2017

Calibration Date